Public Goods and Common Resources

Identify the good → explain the incentives → assess the outcome → evaluate solutions

1. Classify the good: two properties

> **DEFINITION Excludability and rivalry**
>
> A good is **excludable** when it is feasible to prevent someone from using it. A ticket, a locked gate, or a password can make exclusion possible.
>
> A good is **rival in consumption** when one person's use leaves less of it, or reduces its quality, for someone else. If you occupy the only empty seat, nobody else can sit there. When a network becomes congested, an extra user can slow everyone else.

These are different questions. **Can someone be kept out? Does one person's use reduce what others can use?** The answers give four categories.

	Excludable	**Not effectively excludable**
Rival	**Private good.** A reserved cinema seat: entry can be restricted, and a seat taken by one person is unavailable to others.	**Common resource.** Berries in an unfenced woodland where collecting cannot effectively be restricted: berries picked by one visitor are gone for others.
Not rival	**Club good.** A password-protected digital reference library with plenty of capacity: another reader does not interfere with others.	**Public good.** A mural visible from an open public walkway: passersby can see it without permission, and one viewer does not diminish another's view.

> **KEY RESULT The four kinds of goods**
>
> **Private good:** excludable and rival. **Club good:** excludable and nonrival.
> **Common resource:** nonexcludable and rival. **Public good:** nonexcludable and nonrival.

Why does this classification matter? It tells us where private incentives may differ from society's interests. A price of zero does *not* make a good nonexcludable: a museum can offer free admission yet retain the ability to close its gates. Likewise, a crowded free network can be rival. Classification depends on the circumstances, not on who supplies the good or whether a fee is charged. An enforceable patent may restrict commercial use of a design, while knowledge of the design remains nonrival: the legal rule changes excludability, not rivalry.

2. Why might markets fail?

> **DEFINITION Market failure, externality, and free rider**
>
> **Market failure** occurs when private decisions fail to maximize total benefit minus total opportunity cost, also called **total surplus**.
> An **externality** is a benefit or cost of an action that affects others but is not taken into account by the decision maker.
> A **free rider** is someone who receives a benefit without contributing to its cost.

Consider a mural that everyone passing a public walkway can enjoy. If one resident pays to create it, neighbors enjoy the view too, whether they contribute or not. Each resident might hope another will pay. The artist cannot necessarily collect payment from everyone who benefits. **The result can be too little provision, even when the mural is worth more to the community than it costs.**

Now consider berries in open woodland. Each person enjoys what they pick, but may ignore the loss to later visitors or damage to plants from excessive harvesting. **The result can be too much use, because users fail to bear the full cost of their decisions.**

> **KEY RESULT Remember the direction of the problem**
>
> **Public good ⇒ possible underprovision:** the benefits received by others may not be reflected in the supplier's revenue.
> **Common resource ⇒ possible overuse:** the costs imposed on others may not be reflected in the user's decision.
> Neither outcome is inevitable. Cooperation, effective rules, and enforcement can change incentives.

3. Efficient provision of a public good

The economic question is not simply whether people like a public good. It is whether the additional benefit to *everyone together* justifies the resources required to provide it.

> **DEFINITION Marginal benefit and marginal cost**
>
> **Marginal benefit (MB)** is the extra benefit from one more unit. **Marginal cost (MC)** is the extra opportunity cost of supplying it. For a public good, several people enjoy the *same* additional unit, so their marginal benefits must be *added*.

> **KEY RESULT The efficient-quantity rule**
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> For each additional unit of a public good, add the benefits to all users. Provide it when **total marginal benefit is greater than marginal cost**. Stop before the next unit whose total benefit is below its cost. At equality, either choice gives the same net benefit. In the worked example, benefits fall with each clock while cost stays at $100, so stopping before the third clock is efficient.

Worked example: Public clocks for a pedestrian shopping area

Two shopkeepers, A and B, would benefit from large outdoor clocks placed at successive points along a pedestrian shopping route. Customers could see each clock without paying, and one person's reading would not stop anyone else from reading it. **The time display is the public-good service.** For simplicity, assume all benefits in this hypothetical accrue to A and B. The numbers are their maximum willingness to pay—the most each would give up to get *each additional clock*. Each costs $100.

Clock	A's benefit	B's benefit	Total benefit	Cost	Decision
1st	$70	$50	**$120**	$100	Provide
2nd	$62	$43	**$105**	$100	Provide
3rd	$42	$33	**$75**	$100	Stop

For the first clock, A's benefit ($70) plus B's benefit ($50) equals $120. The clock costs $100, so the community gains $20 in net benefit. The second clock adds $105 of benefit at a cost of $100, so it also pays to install. The third costs $100 but adds only $75 of benefit; installing it would waste $25 of resources. **Two clocks maximize total net benefit.**

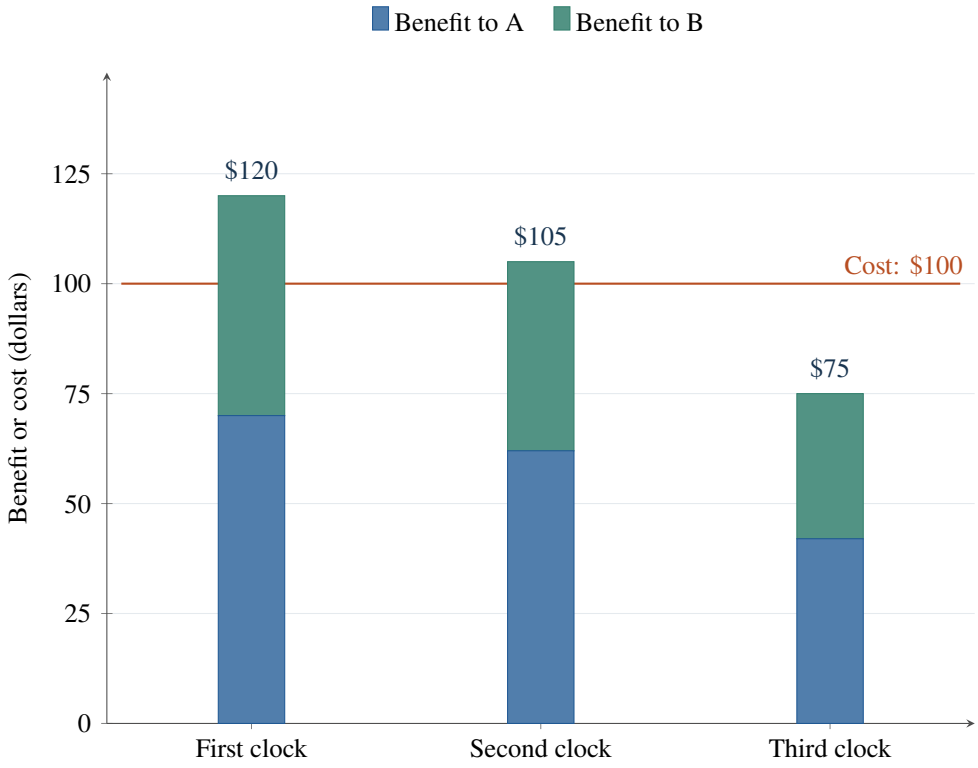


Figure 1. The height of each stacked bar is A's benefit plus B's benefit for that clock. The horizontal line shows its $100 cost. Only the first two bars exceed the cost line.

Read the graph and ask why the market fails. Each colored piece is worth *less* than $100 on its own, but the first two *combined* bars exceed $100. If either shopkeeper must pay the full cost, neither buys a clock; each hopes the other will pay. **Thus the market can supply none even though two clocks increase total surplus.**

KEY RESULT Why might the private market supply too few clocks?

Suppose a clock is installed only if *one shopkeeper pays its entire $100 cost*; there is no way to coordinate shared payment. Neither A nor B will buy even the first clock: A receives only $70 of benefit and B only $50. The same is true of the second clock. **Private provision is zero under this funding arrangement, although two clocks are socially worthwhile.**

Both people could see the clocks after someone else paid, so neither wants to bear the whole cost. That is the **free-rider problem**. If they can agree to split the cost, the failure need not occur. For example, paying $60 by A and $40 by B for each of the first two clocks makes both better off. *Nonexcludability creates the incentive problem; it does not make cooperation impossible.*

Reading a summation sign. Economists sometimes write the total marginal benefit as

$$\underbrace{MB_1 + MB_2 + \cdots + MB_n}_{\text{benefits of all } n \text{ people}} = \sum_{i=1}^{n} MB_i.$$

The symbol $\sum$ means *add*. i labels the people, and n is how many there are. For our two shopkeepers, the first clock gives $MB_A + MB_B = 70 + 50 = 120$. No algebra beyond addition is needed.

4. Common resources and the tragedy of the commons

DEFINITION Tragedy of the commons

The **tragedy of the commons** is the tendency to overuse an open-access, rival resource when users ignore the harm their extra use causes others. The harm is a **negative externality**.

Worked example: Collecting berries in an open woodland

Anyone may collect berries from an unfenced patch. A basket benefits the person who picks it, but extra collecting also leaves fewer berries for later visitors and damages plants. Assume collecting each additional basket costs the picker $3 in time and effort. The numbers below are *hypothetical marginal values*: the benefit and cost of each *additional* basket, not totals for all baskets combined.

Basket	Picker's benefit	Picker's cost	Cost to others	Full social cost
1st	$14	$3	$2	**$5**
2nd	$11	$3	$5	**$8**
3rd	$8	$3	$7	**$10**
4th	$5	$3	$10	**$13**

The private decision. The picker compares personal benefit with the $3 personal cost. Even the fourth basket provides $5 of benefit, so all four are privately worthwhile.

The social decision. The third basket produces $8 of benefit but costs society $3 + $7 = $10. The fourth is still worse. Only the first two baskets provide benefits above their *full* costs. **Private incentives encourage four baskets; the efficient number is two.**

KEY RESULT Private cost is not the whole cost

Social marginal cost = private marginal cost + marginal external cost.

For the third basket, the cost to the picker is \$3 and the harm to other people is \$7, so social marginal cost is \$10. If the picker considers only \$3, the basket will be taken even though its benefit is just \$8. **This gap between private and social cost explains overuse.**

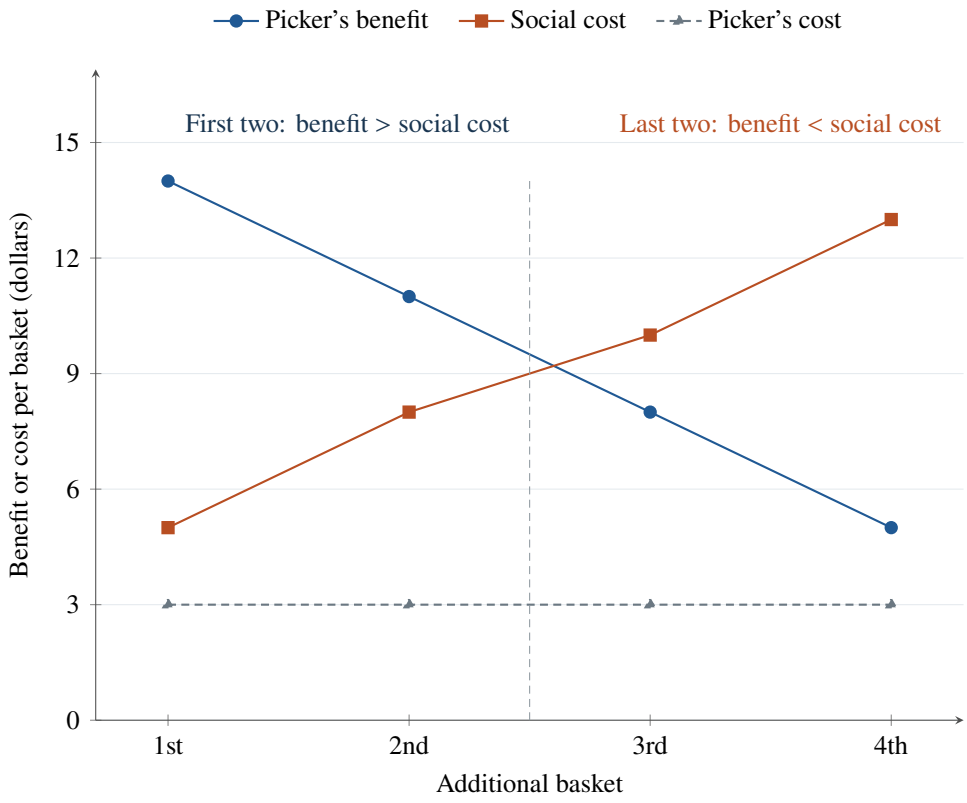


Figure 2. For baskets 3 and 4, the picker's marginal benefit is still above personal cost, but below full social cost. The dashed separator divides worthwhile from excessive baskets; connecting lines are visual guides for discrete units.

Read the graph. The blue line exceeds the gray line for every basket, so the picker wants all four. But the blue line falls below the orange line after basket two, so the last two baskets impose more cost on society than benefit. **The graph shows both why individuals overuse the resource and why society would prefer less use.**

5. How can these failures be addressed?

Begin with the incentive. A public good may lack a means to collect payment from all beneficiaries. A common resource may lack a means to make users bear costs they impose on others. Remedies work by changing those incentives.

KEY RESULT Public good: make worthwhile provision possible

For the clocks, a jointly financed arrangement could fund the first two units, since both shopkeepers benefit after paying their agreed shares. A university or local authority might instead provide a worthwhile service using general revenue. **The economic test remains total benefits versus full costs**; government provision is not automatically efficient.

KEY RESULT Common resource: make users account for scarcity

For the berry patch, an **enforced harvest limit** of two baskets would prevent the excessive third and fourth baskets in the example. Alternatively, a **corrective charge** can make each picker face the costs imposed on others. For instance, a $7 charge on the third basket raises its private cost from $3 to $10. The picker then rejects that basket, because its benefit is only $8.

Effective ownership can give users an incentive to protect future supplies, because they gain from the resource being available later. Monitored community rules can work too. Each remedy needs enforcement and may itself be costly.

Does an access restriction change the type of good? An enforceable permit can make use of the berry patch excludable. *It does not change rivalry:* berries picked by one person remain unavailable to others. An access-controlled rival resource is classified as a private good in this four-category framework.

A useful contrast: positive externalities. Some actions give benefits to other people rather than imposing costs. If the decision maker ignores these extra benefits, there may be *too little* of the activity. What matters is the difference between the private and social benefits or costs.

6. Cost-benefit analysis: is a project worth undertaking?

DEFINITION Cost-benefit analysis

Cost-benefit analysis adds up a project's benefits and compares them with its full **opportunity costs**: the value of resources in their next-best uses. **Net benefit** equals total benefits minus total costs, including relevant effects without market prices.

KEY RESULT The decision rule

For one project, undertake it if **total benefit exceeds total cost**. If choosing between projects that cannot both be undertaken, select the feasible project with the **largest positive net benefit**. If every option has negative net benefit, choose none. Do not confuse a free service for users with a service that costs nothing to provide.

Worked example: Choosing between two improvements

A town can afford only one of two improvements. A shaded waiting area would produce estimated benefits of \$16,000 and cost \$12,000, giving net benefit \$4,000. A new outdoor seating area would produce benefits of \$13,000 and cost \$7,000, giving net benefit \$6,000. **Choose the seating area.** Its benefits are smaller, but its *net* benefit is greater. The figures are illustrative estimates, not observed data.

Why is valuation difficult? People may benefit from a service without buying it, so no ordinary price reveals its value. Survey answers can be biased if respondents expect them to influence what they receive or pay.

> **DEFINITION Value of a statistical life (VSL)**
>
> Economists sometimes need to evaluate safety measures. A **value of a statistical life (VSL)** summarizes people's willingness to pay for small reductions in their risk of death. It does *not* put a price on a particular person's life.
>
> One way to estimate VSL is to compare otherwise similar jobs that have different fatality risks and pay. If a worker requires an extra \$400 each year to accept an additional annual death risk of 1 in 20,000 (0.00005), the implied VSL is \$400 divided by 0.00005, or \$8 million. This illustrates the calculation; actual estimates require careful comparison of jobs and workers.

Worked example: A hypothetical safety improvement

Suppose a safety project lowers the probability of a fatal accident by *one in 10,000* (0.0001) for each of 1,000 people over one year. Using an illustrative VSL of \$8 million, the total expected benefit is

$$\underbrace{1{,}000 \times 0.0001}_{\text{0.1 expected deaths avoided}} \times \$8{,}000{,}000 = \boxed{\$800{,}000}.$$

Expected deaths avoided means a statistical average, not a fraction of an actual person. If the project's total cost for the same period is \$600,000, its estimated net benefit is \$200,000. **The lesson is to compare the value of risk reduction with its opportunity cost.** The number 0.0001 is a probability, equivalent to 0.01%, not 1%.

7. Why might an efficient project lose a vote?

> **KEY RESULT Voting and efficiency ask different questions**
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> A voter favors a proposal when *their own benefit* exceeds *their own tax payment*. A majority vote counts supporters.
>
> Economic efficiency adds benefits and costs across *everybody*. A project may therefore be worthwhile for society even though most individuals vote against it. The reverse can also happen.

Worked example: three residents. A shared improvement costs \$90, with a \$30 tax on each of three residents. A values it at \$55, B at \$25, and C at \$20. Total benefit is \$100, so net social benefit is **+\$10**. Yet only A votes yes: B and C each value the project below their own \$30 tax. **It loses 2–1 despite being efficient.** A vote counts individual gains; cost-benefit analysis adds everyone's gains and costs.